

Project
Design Layout and Analysis of a 2-stage
Op-Amp

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1 OBJECTIVE OF THE EXPERIMENT

1.1 Group 2 Specifications

General Specifications:

- Supply Voltage (V_{DD}): 1.8 V
- Reference Current Source (I_{ref}): 20 μ A
- Slew Rate: 1V/us
- Phase Margin: $\geq 60^\circ$
- Load Capacitance (C_L): 10 pF
- Input Common Mode Range (ICMR): 0.6 V to 1.4 V

Target: High Gain

Design Requirements:

- DC Gain: ≥ 75 dB
- Gain Bandwidth Product (GBW): ≥ 10 MHz
- Power Dissipation (P_{diss}): ≤ 1 mW
- Maximum Channel Length (L_{max}): ≤ 2 μ m

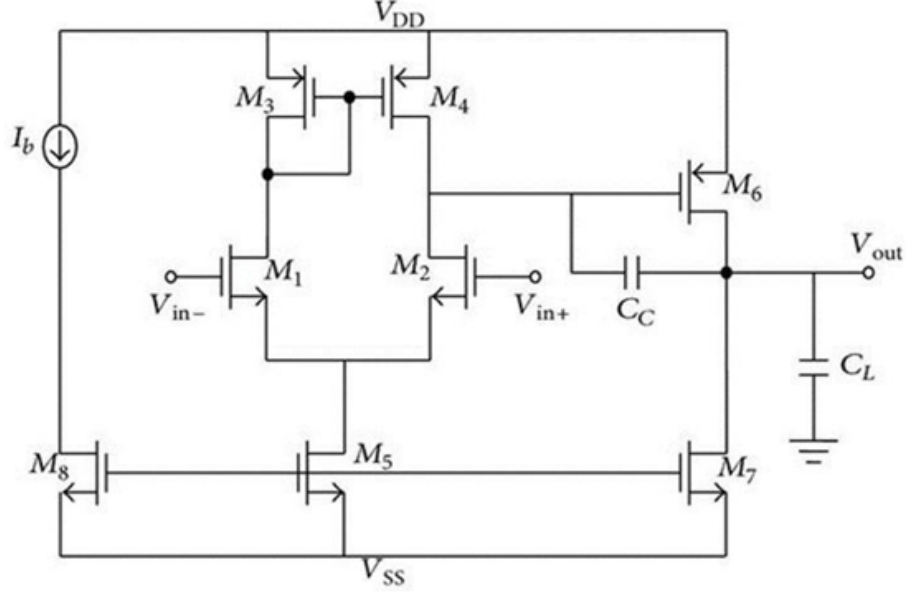


Figure 1: Circuit Diagram of 2 stage op-amp

2 g_m/I_D METHOD

2.1 Limitations of Conventional Analog Design Methods

Traditional analog CMOS design techniques are commonly derived from the square-law model of the MOSFET, where the drain current (I_D) is expressed as a function of gate overdrive voltage ($V_{GS} - V_{TH}$). Although this approximation is useful for basic understanding, it becomes increasingly inaccurate for modern deep-submicron technologies.

Several physical effects are not properly captured by the ideal square-law model, such as:

- Drain Induced Barrier Lowering (DIBL) and Reverse Short Channel Effects
- Variation of threshold voltage with channel dimensions
- Channel Length Modulation

- Mobility degradation at high electric fields
- Velocity saturation in short-channel devices
- Process and temperature variations

Because of these non-idealities, hand calculations based only on classical equations often differ significantly from simulation and fabricated circuit behavior. This motivates the use of a more practical and technology-aware design methodology.

The g_m/I_D approach overcomes these issues by relying on transistor characterization data obtained from simulation models. It provides improved agreement between theoretical sizing calculations and actual circuit performance.

2.2 Why the g_m/I_D Method is Preferred

The ratio g_m/I_D represents the transconductance generated per unit bias current. In simple terms, it measures how efficiently a transistor converts current (power) into gain.

This makes the method highly valuable in modern analog design, especially where power efficiency and performance must be balanced.

- Higher g_m/I_D implies better gain for a given current consumption.
- It is based on accurate device models and therefore reflects real CMOS behavior.
- It is highly suitable for low-power and battery-operated circuits.
- It helps in designing high-speed amplifiers with bandwidth constraints.
- It supports noise-performance optimization.
- It is valid across all inversion regions:
 - Weak inversion
 - Moderate inversion
 - Strong inversion
- It simplifies iterative transistor sizing using lookup plots or tables.

- It offers a reusable and structured design flow for future circuits.

In practical design, a suitable g_m/I_D value is selected depending on whether the objective is high speed, low noise, low power, or high gain. The corresponding transistor dimensions and bias conditions are then extracted from pre-generated charts.

2.3 Preparation of g_m/I_D Design Charts

The g_m/I_D methodology uses several important transistor performance metrics, which are extracted through DC and AC simulations.

- **Transit Frequency**

$$f_T = \frac{g_m}{2\pi C_{gs}}$$

This indicates the maximum useful frequency or bandwidth capability of the device.

- **Intrinsic Gain**

$$g_m r_o = \frac{g_m}{g_{ds}}$$

This gives the maximum achievable voltage gain of a single transistor.

- **Transconductance Efficiency**

$$\frac{g_m}{I_D}$$

This shows how effectively bias current is converted into transconductance.

To build the lookup database, the MOS transistor is simulated over a wide operating range and the following plots are generated:

1. g_m versus V_{GS}
2. I_D versus V_{GS}
3. g_m/I_D versus V_{GS}
4. f_T versus V_{GS}

5. g_m/g_{ds} versus V_{GS}

From these primary curves, secondary design charts are created such as:

- f_T versus g_m/I_D
- g_m/g_{ds} versus g_m/I_D
- I_D/W versus g_m/I_D

These charts allow designers to directly choose transistor width, operating region, current density, gain, and speed without repeatedly solving nonlinear equations.

2.4 Advantages in Practical Analog Design

Compared with classical hand analysis, the g_m/I_D method provides:

- Faster transistor sizing
- Better correlation with SPICE simulation
- Easier optimization of power versus speed
- Reliable design across technology nodes
- Greater insight into inversion-region based operation

Hence, the g_m/I_D methodology has become one of the most widely used modern techniques for CMOS analog and mixed-signal circuit design.

2.5 Application of g_m/I_D Charts in Circuit Design

One of the major advantages of the g_m/I_D design methodology is that the ratio g_m/I_D is almost independent of transistor aspect ratio (W/L) for positive gate overdrive voltage:

$$V_{OV} = V_{GS} - V_{TH} > 0$$

This property makes the method highly practical for transistor sizing. In addition, the variation of g_m/I_D across different CMOS technologies is relatively small, allowing the methodology to remain useful even when process nodes change.

The practical range of g_m/I_D generally lies between:

$$5 \leq \frac{g_m}{I_D} \leq 25$$

Lower values correspond to strong inversion operation, whereas higher values indicate weak or moderate inversion operation.

2.5.1 Guidelines for Selecting g_m/I_D

While assigning g_m/I_D values to MOS devices in a circuit, the following design rules are commonly followed:

1. **Low g_m/I_D values** are preferred for transistors whose transconductance does not significantly affect gain, such as active loads or bias devices. These values generally provide:
 - Smaller transistor area
 - Higher speed
 - Lower parasitic capacitances
2. **High g_m/I_D values** are selected for devices that strongly influence amplifier gain, such as differential input transistors. These values offer:
 - Higher gain efficiency
 - Lower flicker noise
 - Reduced mismatch
 - Larger signal swing
 - Better low-power performance

2.5.2 Step-by-Step Design Procedure

After choosing an appropriate g_m/I_D ratio for each transistor, the remaining design parameters are determined using lookup charts.

1. Select drain current I_D based on the available power budget.
2. Choose channel length L such that the intrinsic gain:

$$\frac{g_m}{g_{ds}}$$

is sufficient to meet the overall gain requirement.

3. Verify that the selected channel length does not excessively reduce transit frequency f_T , since high gain and high bandwidth must be balanced carefully.
4. Obtain transistor width W using the chart of:

$$\frac{I_D}{W} \text{ versus } \frac{g_m}{I_D}$$

5. Determine the required DC operating voltage from the:

$$\frac{g_m}{I_D} \text{ versus } V_{OV}$$

characteristic.

2.5.3 Important Design Trade-Off

A very high intrinsic gain can be achieved by using longer channel devices. However, this often reduces transistor speed and lowers bandwidth. Therefore, practical analog design always requires balancing:

- Gain
- Power consumption
- Bandwidth
- Noise
- Silicon area
- Stability

2.5.4 Summary

The g_m/I_D technique transforms analog design into a structured optimization process. Instead of relying solely on approximate equations, designers can directly use simulation-generated charts to choose transistor dimensions and operating points with better accuracy and faster convergence.

3 Design of Two-Stage Op-Amp using g_m/I_D Method

3.1 Miller Compensation Capacitor and Input Transconductance Calculation

To ensure sufficient phase margin and to implement pole splitting for frequency compensation, the compensation capacitor C_c must satisfy:

$$C_c > 0.22 \cdot C_L$$

Where:

- $C_L = 10 \text{ pF}$ is the load capacitance.

Substituting the values:

$$C_c > 0.22 \cdot 10 \text{ pF} = 2.2 \text{ pF}$$

To provide margin, we select:

$$C_c = 4 \text{ pF}$$

3.2 Input Transconductance g_m

The gain-bandwidth product (GBW) constraint dictates the minimum required input stage transconductance:

$$\text{GBW} = \frac{g_m}{2\pi C_c}$$

Given that the required gain bandwidth is:

$$\text{GBW} \geq 10\text{MHz}$$

Substituting $C_c = 4\text{pF}$:

$$\begin{aligned} g_m &\geq 2\pi \cdot C_c \cdot \text{GBW} \\ g_m &\geq 2\pi \cdot 4 \times 10^{-12} \cdot 10 \times 10^6 \\ g_m &\approx 188.6 \mu\text{S} \end{aligned}$$

I selected less g_m than theoretical value, because we have to use resistor to get GBW in my design,

$$g_m = 188.6 \mu\text{S}$$

This g_m value corresponds to the input differential pair formed by transistors M1 and M2.

3.3 Summary

- Load Capacitance $C_L = 10\text{pF}$
- Selected Miller Cap $C_c = 4\text{pF}$
- Gain Bandwidth Requirement $\text{GBW} \geq 10\text{MHz}$
- Required Transconductance $g_m = 188.6 \mu\text{S}$

3.4 Selection of g_m/I_D for Different Transistors

The design of a two-stage operational amplifier using the g_m/I_D methodology begins with selecting appropriate g_m/I_D values for the transistors based on their functional requirements.

Transistors that require a large signal swing must operate in a region of higher intrinsic gain, which corresponds to a larger g_m/I_D value. On the other hand, transistors primarily used for biasing or current mirroring can function efficiently with a smaller g_m/I_D , prioritizing speed and area.

- **Input NMOS transistors (M1, M2):** A large g_m/I_D value of **11** is chosen for M1 and M2 because their transconductance contributes directly to the gain of the differential pair. A higher g_m/I_D is desirable in this case to ensure low flicker noise, less mismatch, and larger voltage swings, all of which improve the overall performance of the amplifier.
- **Current Mirror and Load Transistors (M3, M4, M5, M6):** A smaller g_m/I_D value of **10** is chosen for these transistors. Since their transconductance does not significantly contribute to the gain and their primary function is current copying or load operations, a lower g_m/I_D is sufficient. This choice helps in reducing area and improving speed without sacrificing performance.

3.5 Preparation for Plotting: Extraction of NMOS and PMOS Operating Point Data

For accurate generation of g_m/I_D design curves, transistor operating point parameters must first be extracted from simulation. These parameters are later used to construct lookup charts such as g_m/I_D versus V_{GS} , I_D/W versus g_m/I_D , intrinsic gain plots, and transit frequency curves.

The first step is to perform DC characterization of both NMOS and PMOS devices while ensuring that all device operating point values are saved during simulation.

3.6 NMOS Testbench Setup in Cadence

To characterize the NMOS transistor, a dedicated testbench is created in Cadence Virtuoso.

1. Place a single NMOS transistor with appropriate bias connections.
2. Connect the drain and gate terminals to suitable DC voltage sources.
3. Keep the source and bulk terminals connected to ground unless body-bias analysis is required.
4. Choose transistor dimensions according to the technology minimum or desired channel length.
5. Configure a DC sweep for V_{GS} over the intended operating range.

6. Run the simulation while enabling operating point data storage.

The DC sweep allows observation of the drain current variation with gate voltage and helps identify weak, moderate, and strong inversion regions.

3.7 Creating a Save File for Operating Parameters

To store all relevant transistor operating point parameters, a Spectre save file must be created.

1. Open the terminal and create a new file with `.scs` extension:

```
touch nmos_1.scs
```

2. Edit the file using a text editor:

```
nano nmos_1.scs
```

3. Insert the following command inside the file:

```
save NM0:all
```

4. Save the file and exit the editor.

Here, NM0 represents the instance name of the NMOS transistor used in the schematic. If a different name is used, it must be updated accordingly.

The command `:all` ensures that all available operating point parameters are saved, including:

- Drain current (I_D)
- Transconductance (g_m)
- Output conductance (g_{ds})
- Threshold voltage (V_{TH})
- Gate-source voltage (V_{GS})
- Capacitances such as C_{gs} and C_{gd}

3.8 Including the Save File in ADE XL

After creating the save file, it must be linked with the simulation environment.

1. Open ADE_XL or ADE Explorer.
2. Go to:
Setup → Simulation Files
3. Under the Definition Files section, click Add.
4. Browse and select the file `nmos_1.scs`.
5. Click Apply and save the setup.

Once linked, every simulation run will automatically store the specified operating point data.

3.9 Importance of Saved Operating Points

The extracted data can later be used to generate important design charts such as:

- g_m/I_D versus V_{GS}
- I_D/W versus g_m/I_D
- g_m/g_{ds} versus g_m/I_D
- f_T versus g_m/I_D
- V_{OV} versus g_m/I_D

These plots form the foundation of the g_m/I_D based analog design methodology and enable accurate transistor sizing.

3.10 Extension to PMOS Characterization

The same procedure is repeated for PMOS devices using a PMOS testbench and corresponding instance save command such as:

```
save PM0:all
```

This allows complete characterization of both transistor types required for CMOS analog circuit design.

3.11 Fixing the Channel Length using g_m/g_{ds} vs g_m/I_D

After running the DC sweep and saving all operating points, open the **Results Browser** in Cadence. Click on the DC analysis result to view all the saved signal quantities.

To fix the length L of the transistor, we use the plot of g_m/g_{ds} versus g_m/I_D . The quantity g_m/g_{ds} is representative of the intrinsic gain of the device, combining both transconductance and output resistance:

$$\frac{g_m}{g_{ds}} = \text{gain}$$

For this design, a total gain of approximately 75 dB is required. To ensure margin, we target a gain of 76dB.

3.12 Plotting in Cadence

1. In the Results Browser, select the signals: `gm`, `gds`, and `gmoverid`.
2. Send each of these to the calculator.
3. In the calculator, use the expression:

$$\frac{g_m}{g_{ds}} \text{ vs } \frac{g_m}{I_D}$$

using the `waveform vs waveform` function.

4. Parametrically vary the channel length L and generate the plot.

3.13 g_m/I_D Plots

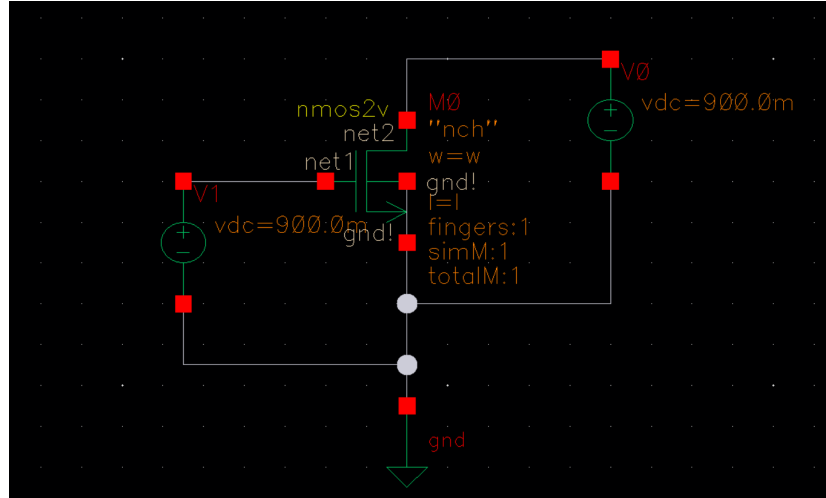


Figure 2: NMOS characterization testbench used for DC sweep analysis.

A diode-connected NMOS transistor is biased using a DC voltage sweep to extract device parameters over all inversion regions. This setup is commonly used to generate lookup plots for the g_m/I_D design method.

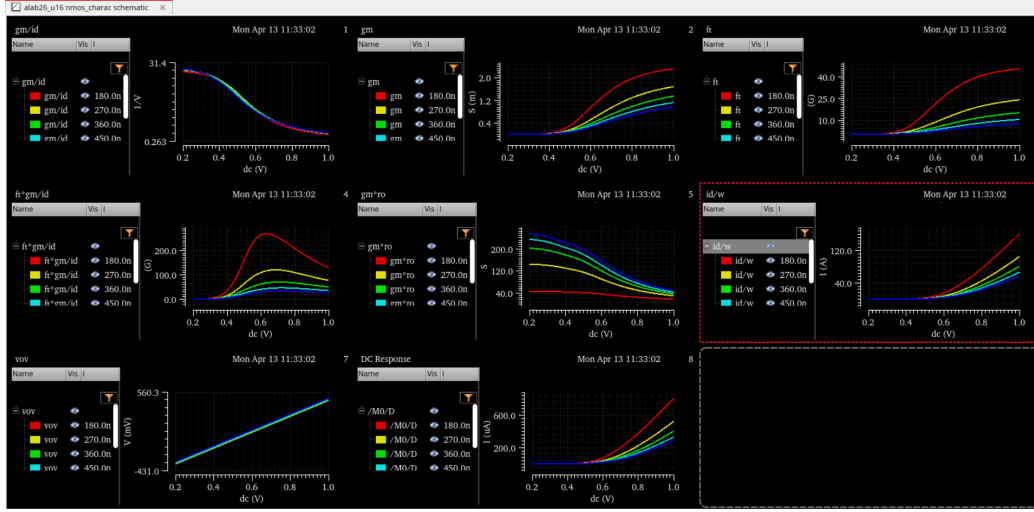


Figure 3: Complete set of NMOS characterization plots.

Multiple transistor metrics are plotted simultaneously against gate voltage for different channel lengths. These curves help designers choose the optimum operating point based on gain, speed, and power trade-offs.

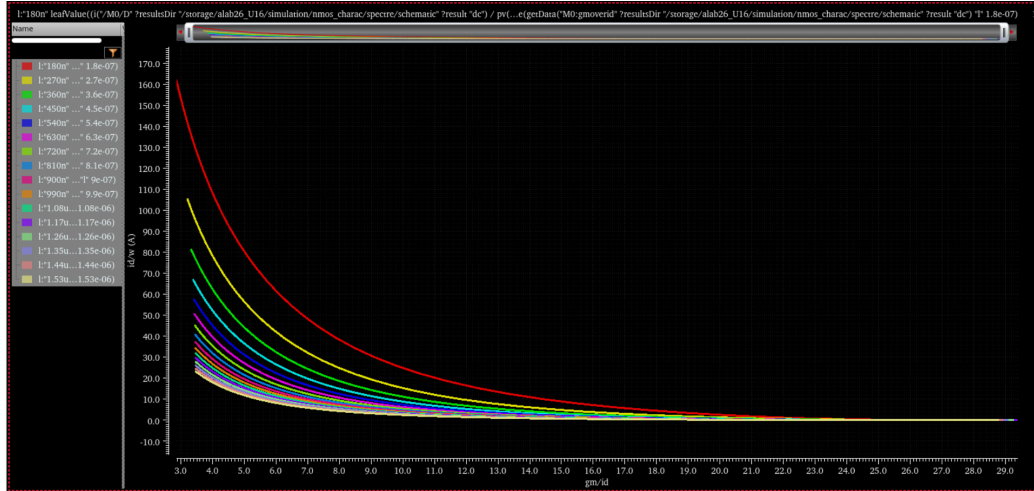


Figure 4: I_D/W versus g_m/I_D for multiple channel lengths.

Current density decreases as g_m/I_D increases, showing that high efficiency requires larger transistor area. Longer channel devices need lower current density for the same inversion level.

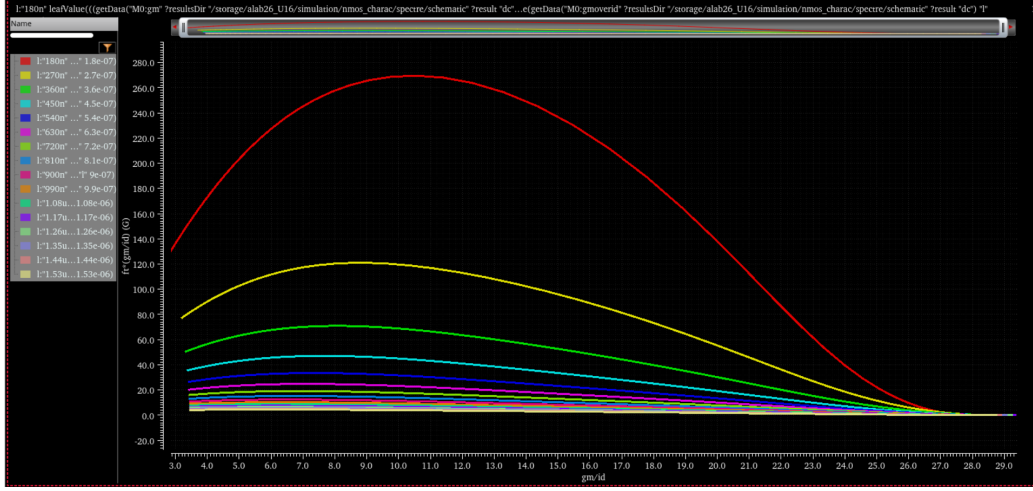


Figure 5: $f_T(g_m/I_D)$ versus g_m/I_D for different channel lengths.

This combined metric indicates the optimum trade-off between speed and efficiency. Shorter channel devices achieve higher peak values due to improved transit frequency.

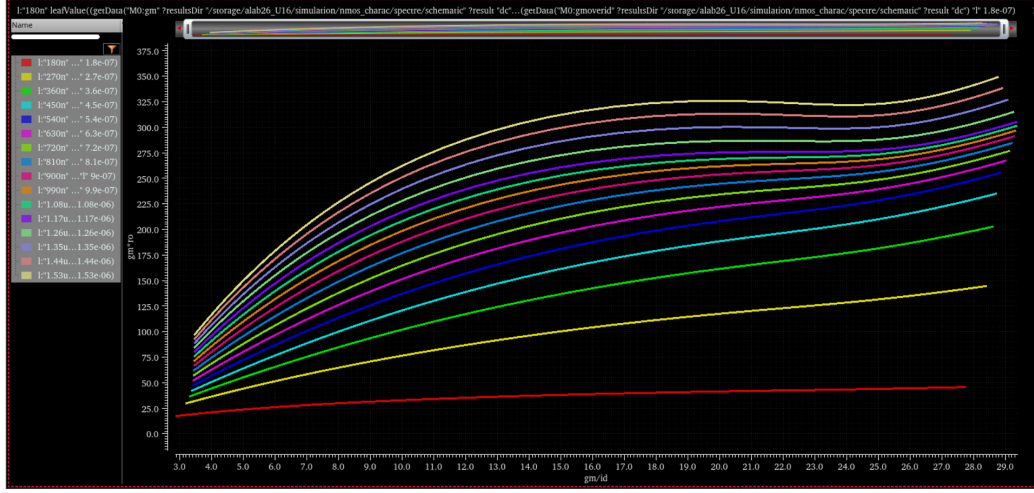


Figure 6: $g_m r_o$ versus g_m/I_D for multiple channel lengths.

Intrinsic gain increases significantly with longer channel length because output resistance becomes larger. This plot is useful for selecting devices in high-gain amplifier stages.

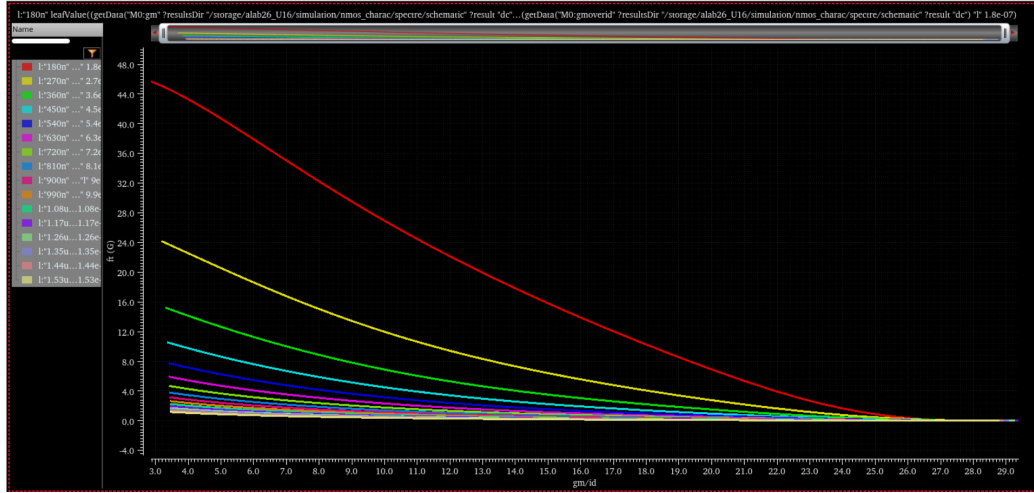


Figure 7: f_T versus g_m/I_D for multiple channel lengths.

As g_m/I_D increases, the device becomes more current-efficient but slower,

causing f_T to decrease. Short-channel transistors provide higher bandwidth than long-channel devices.

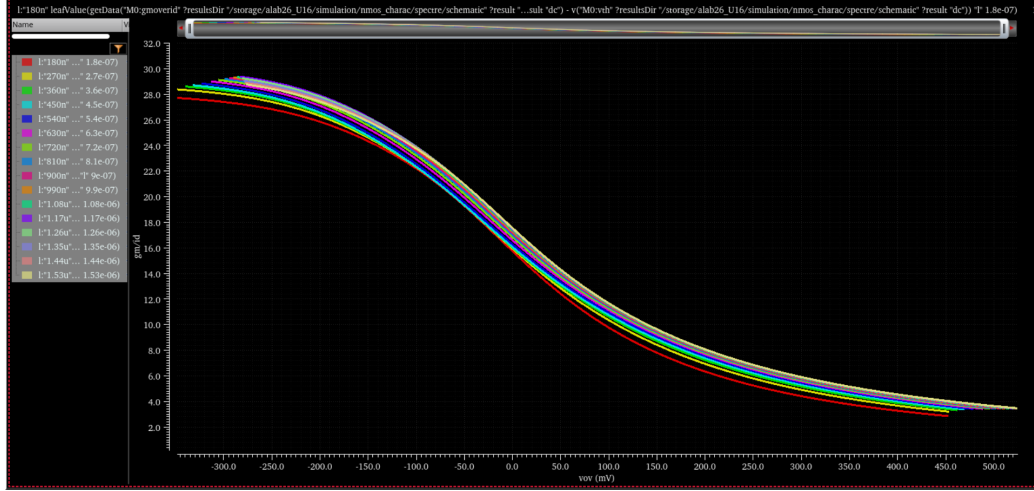


Figure 8: g_m/I_D versus V_{OV} for different channel lengths.

The transconductance efficiency decreases with increasing overdrive voltage. Lower V_{OV} operation corresponds to weak inversion and better power efficiency.

3.14 PMOS Characterization

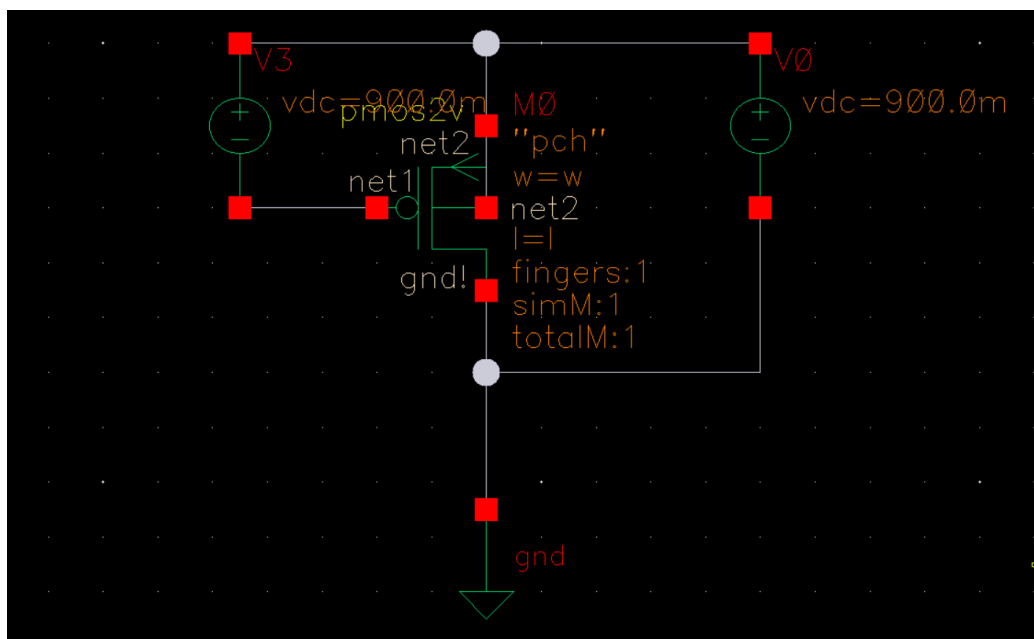


Figure 9: PMOS transistor characterization schematic used for DC sweep analysis.

The PMOS transistor is biased with fixed source potential while the gate voltage is swept to obtain device characteristics for multiple channel lengths.

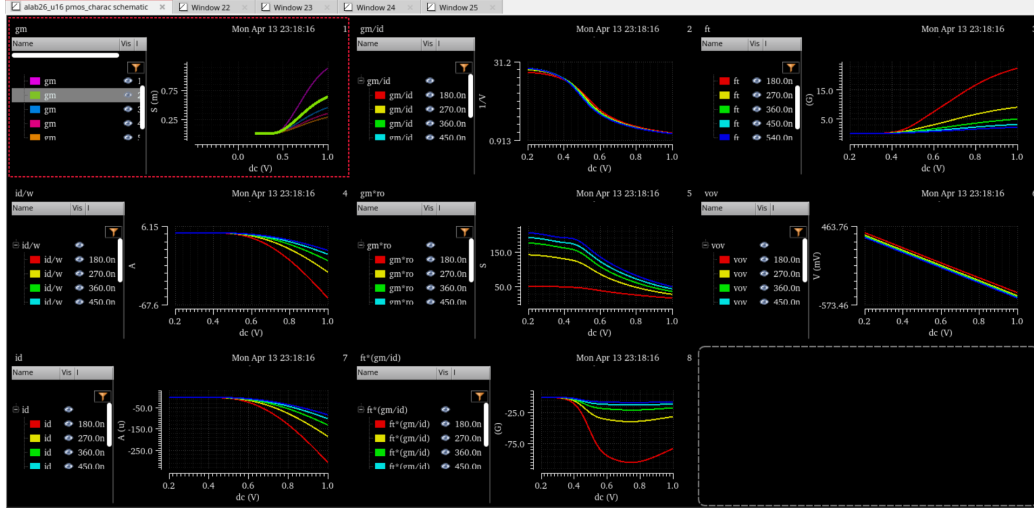


Figure 10: Combined PMOS transistor characterization plots for different channel lengths.

This figure summarizes extracted parameters such as g_m , g_m/I_D , f_T , I_D/W , $g_m r_o$, V_{OV} , drain current, and $f_T(g_m/I_D)$.

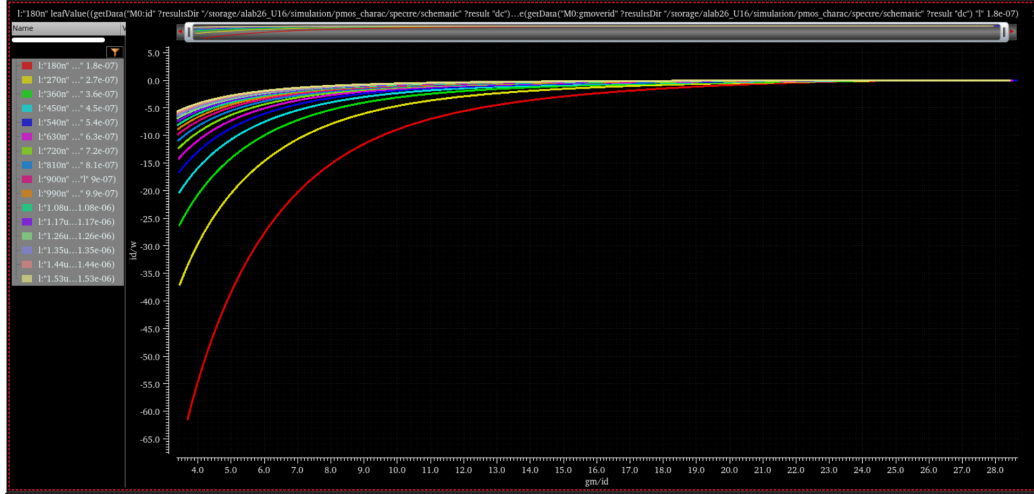


Figure 11: I_D/W versus g_m/I_D for PMOS transistor.

Current density decreases as g_m/I_D increases, indicating higher transconductance efficiency at lower current operation.

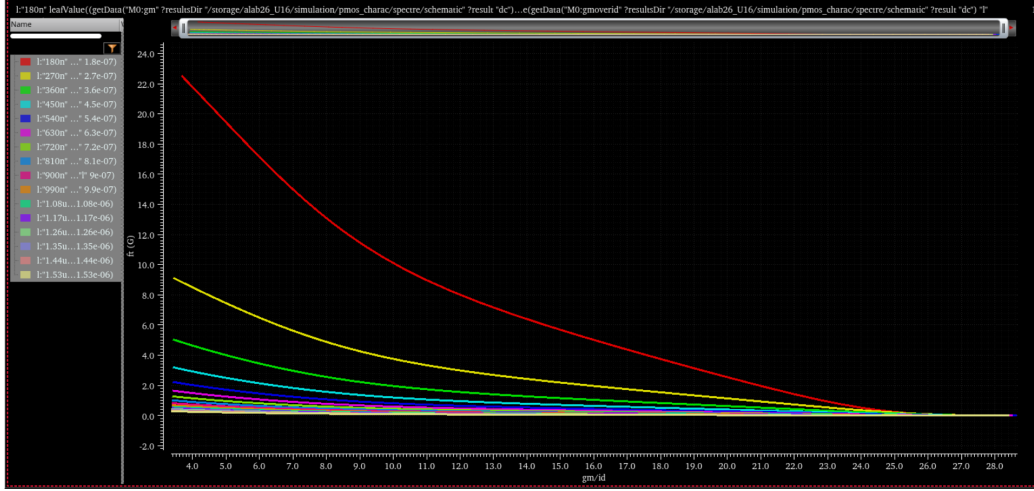


Figure 12: f_T versus g_m/I_D for PMOS transistor.

Transit frequency decreases with increasing g_m/I_D , showing the trade-off between speed and efficiency.

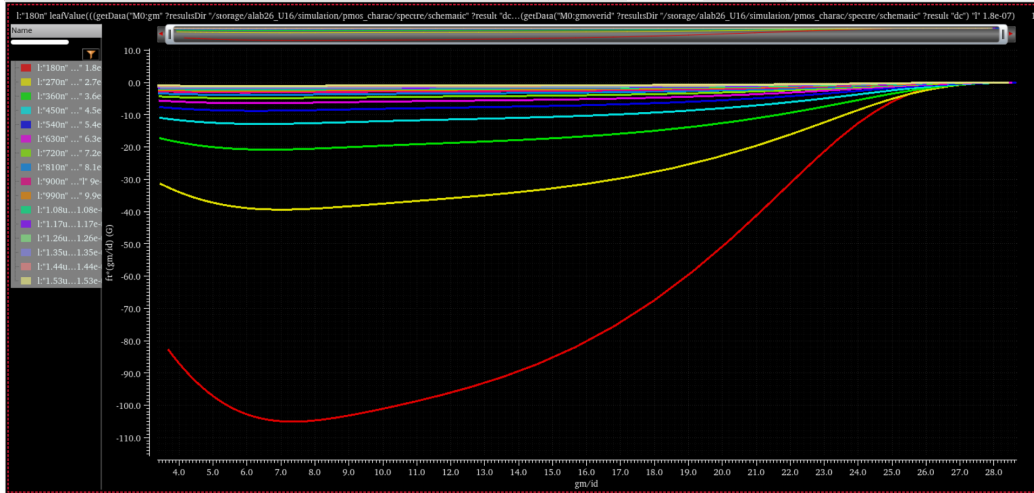


Figure 13: $f_T(g_m/I_D)$ versus g_m/I_D for PMOS transistor.

Short-channel PMOS devices provide higher peak values and better high-frequency performance.

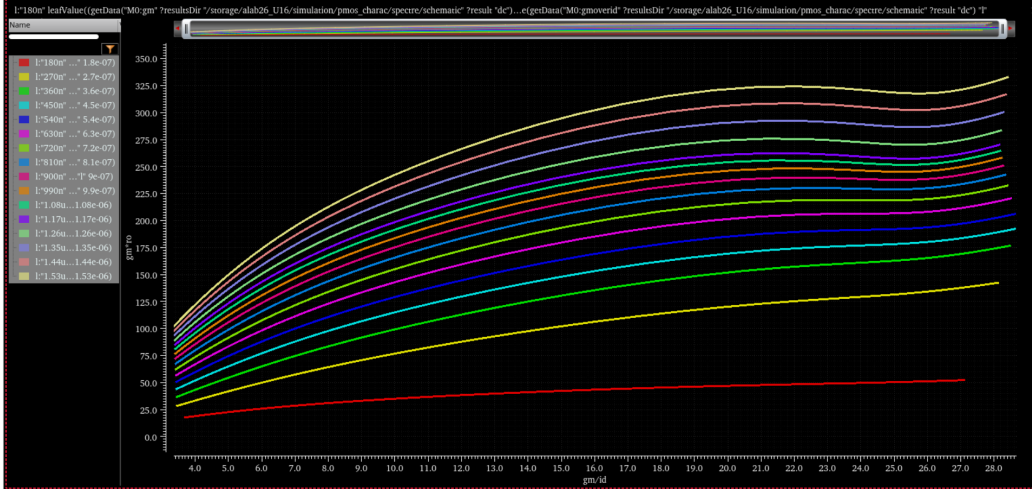


Figure 14: $g_m r_o$ versus g_m/I_D for PMOS transistor.

Intrinsic gain increases with channel length due to larger output resistance.



Figure 15: g_m/I_D versus V_{OV} for PMOS transistor.

Higher g_m/I_D corresponds to lower magnitude overdrive voltage, representing weak inversion operation. This circuit provides the required source bias and swept gate voltage to extract the PMOS design curves used in g_m/I_D based analog design.

4 Design of a Two-Stage Operational Amplifier – Hand Calculations

The design of the two-stage operational amplifier involves determining transistor dimensions using the g_m/I_D methodology to satisfy gain, bandwidth, slew-rate, and stability requirements.

4.1 Design of Transistors M1 and M2

Given:

$$C_L = 10pF, \quad C_C = 4pF$$

The required transconductance is:

$$g_m = 10 \times 2\pi \times C_C$$

$$g_m = 10^7 \times 2\pi \times 4 \times 10^{-12}$$

$$g_m = 188.55 \mu S$$

For M1 and M2:

$$L = 1080nm = 1.08\mu m$$

$$\frac{g_m}{I_D} = 13.604 \quad at V_{ov} = 60mV$$

Hence,

$$I_D = \frac{g_m}{g_m/I_D}$$

$$I_D = \frac{188.55}{13.604}$$

$$I_D = 13.86 \mu A$$

From lookup chart:

$$\frac{I_D}{W} = 1.427$$

$$W = \frac{13.86}{1.427} = 9.712 \mu m$$

Therefore, for M1 and M2:

$$W = 9.712 \mu m, \quad L = 1.08 \mu m$$

4.2 Design of PMOS Load Transistors M3 and M4

Given:

$$L = 1.44 \mu m$$

$$\frac{g_m}{I_D} = 6.136 \quad at V_{ov} = 250 mV$$

From chart:

$$\frac{I_D}{W} = 1.942$$

Using the same branch current:

$$W = \frac{13.86}{1.942} = 7.136 \mu m$$

Thus:

$$W = 7.136 \mu m, \quad L = 1.44 \mu m$$

4.3 Design of Tail Current Source M5

Given:

$$L = 900nm = 0.9\mu m$$

$$\frac{g_m}{I_D} = 13.4439 \quad at V_{ov} = 60mV$$

Tail current:

$$I_D = 2 \times 13.86 = 27.72\mu A$$

From chart:

$$\frac{I_D}{W} = 1.79$$

$$W = \frac{27.72}{1.79} = 15.486\mu m$$

Therefore:

$$W = 15.486\mu m, \quad L = 0.9\mu m$$

4.4 Design of Second Stage Transistor M6

Given:

$$L = 720nm = 0.72\mu m$$

$$\frac{g_m}{I_D} = 6.24 \quad at V_{ov} = 250mV$$

From calculation:

$$g_{m6} = 1.3 \times 1.885mS$$

$$g_{m6} = 2.45mS$$

Thus:

$$I_D = \frac{g_m}{g_m/I_D}$$

$$I_D = \frac{2.45m}{6.24} = 392.63\mu A$$

From chart:

$$\frac{I_D}{W} = 4.02$$

$$W = \frac{392.63}{4.02} = 97.67\mu m$$

Hence:

$$W = 97.67\mu m, \quad L = 0.72\mu m$$

4.5 Design of Current Mirror Transistor M7

Given:

$$L = 900nm = 0.9\mu m$$

$$\frac{g_m}{I_D} = 13.44 \quad at V_{ov} = 60mV$$

$$\frac{I_D}{W} = 1.79$$

$$W = \frac{392.63}{1.79} = 208.34\mu m$$

Thus:

$$W = 208.34\mu m, \quad L = 0.9\mu m$$

4.6 Design of Bias Transistor M8

Given:

$$L = 900nm = 0.9\mu m$$

$$W = \frac{20}{1.79} = 11.173\mu m$$

Hence:

$$W = 11.173\mu m, \quad L = 0.9\mu m$$

4.7 Final Transistor Dimensions

Transistor	Width (μm)	Length (μm)
M1	9.712	1.08
M2	9.712	1.08
M3	7.136	1.44
M4	7.136	1.44
M5	15.486	0.90
M6	97.67	0.72
M7	208.34	0.90
M8	11.173	0.90

Table 1: Final MOS Transistor Dimensions

4.8 Current Distribution Based on Power and Slew Rate Constraints

In this section, we determine the current allocation in the operational amplifier based on the overall power budget and dynamic performance requirements.

4.9 Power Constraint

The total power consumption is constrained by:

$$P \leq 1 \text{ mW}$$

Since power is related to the supply voltage and current as:

$$P = V_{DD} \cdot I_D$$

Given $V_{DD} = 1.8 \text{ V}$, we can calculate the maximum allowable current:

$$I_D \leq \frac{1 \text{ mW}}{1.8 \text{ V}} \approx 555 \mu\text{A}$$

4.10 Summary of Current Distribution

Transistor(s)	Current (μA)
M7 (Tail Current Source)	392.63
M5 (Tail Current Source)	27.76
M1, M2 (Input Pair)	13.86 each
M3, M4 (Current Mirrors)	13.86 each

5 Two-Stage Operational Amplifier

A two-stage operational amplifier is widely used to achieve high voltage gain, large output swing, and improved driving capability in analog integrated circuits. It consists of a differential input gain stage followed by a second common-source gain stage, while Miller compensation is used to ensure frequency stability.

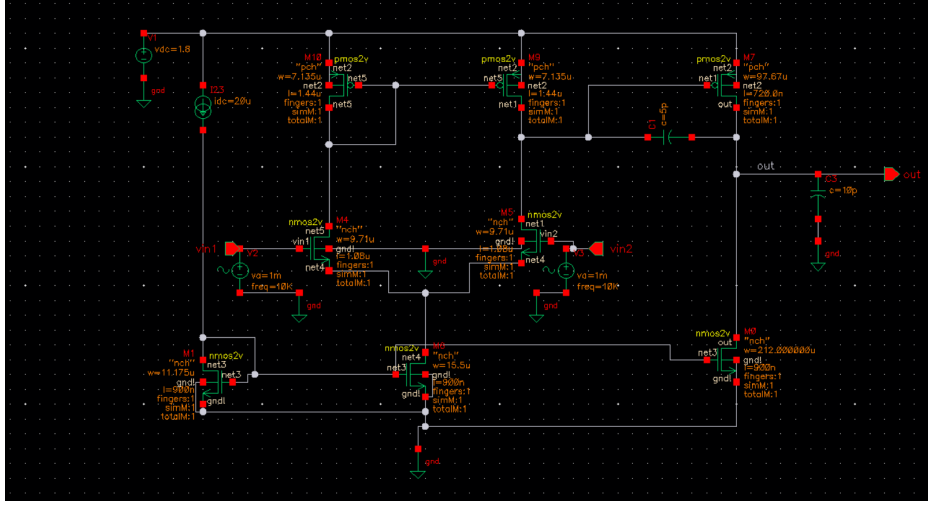


Figure 16: Two-stage CMOS operational amplifier schematic with transistor Sizes.

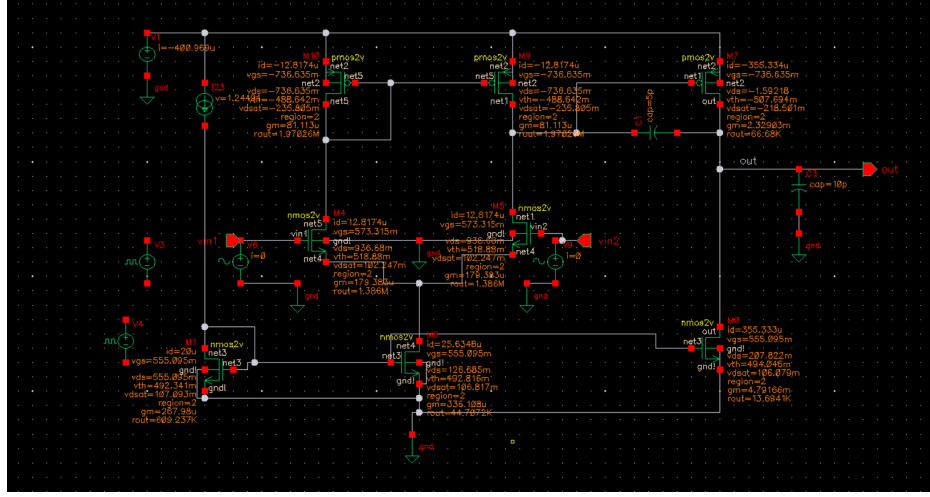


Figure 17: DC operating points at $V_{in}=0.7V$.

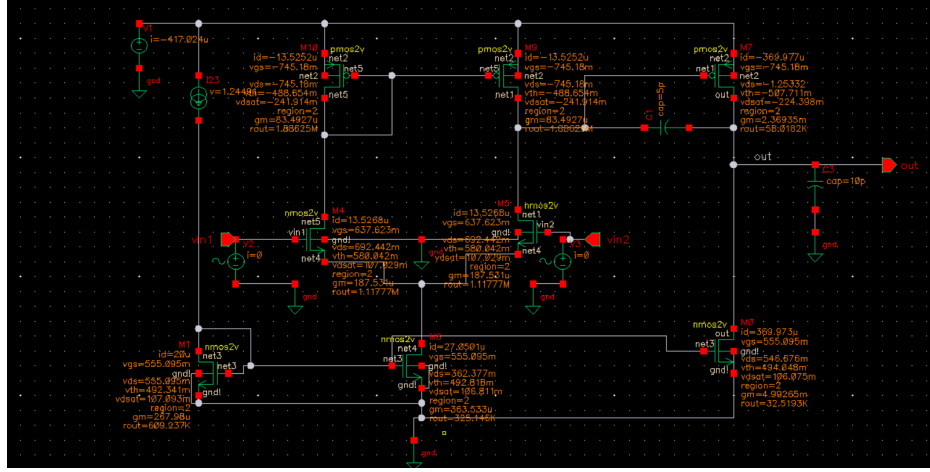


Figure 18: DC operating points at $V_{in}=1V$.

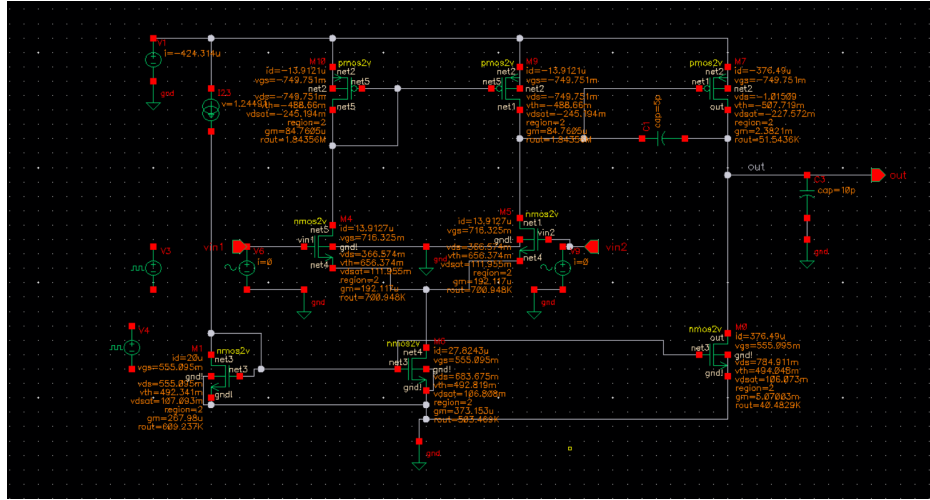


Figure 19: DC operating points at $V_{in}=1.4V$.

5.1 Power calculation

Observed value of average total current= $424.31\mu A$

So the power is $1.8 \times 424.31\mu = 0.7632 \text{ mW}$

5.2 AC analysis before layout:

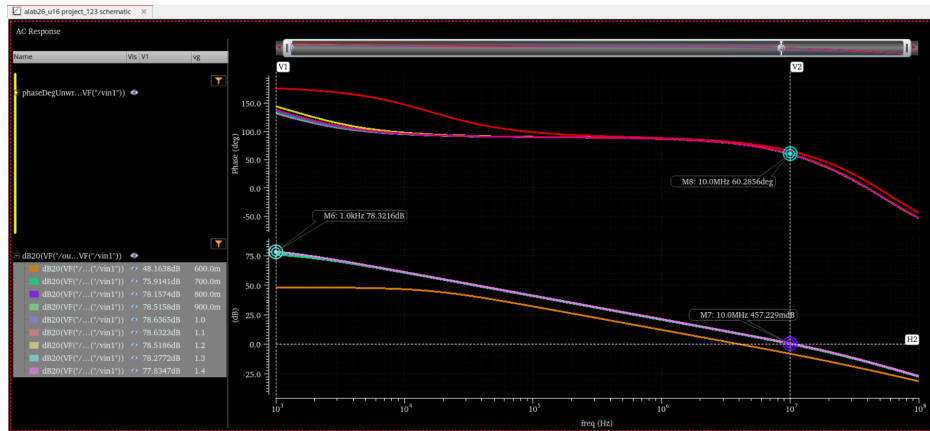


Figure 20: AC response of the amplifier showing gain and phase characteristics. For input voltage range from 0.6V to 1.4V

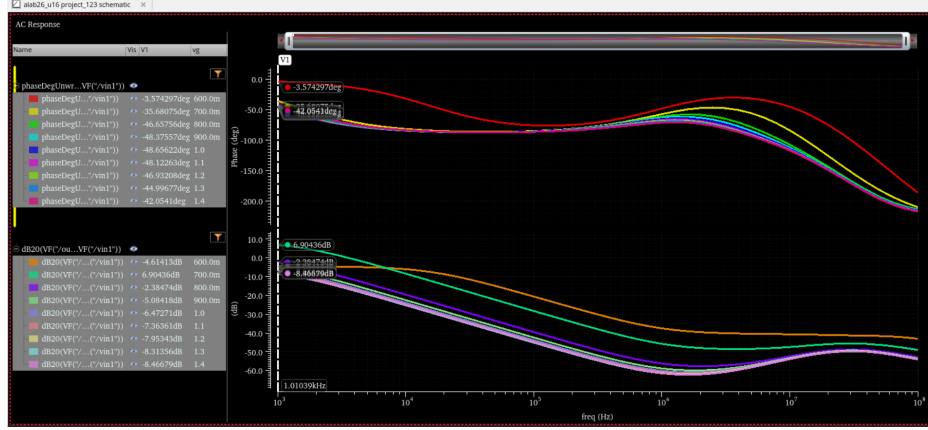


Figure 21: AC response of the amplifier showing common mode gain and phase characteristics. For input voltage range from 0.6V to 1.4V

Table 2: Differential Gain, Common-Mode Gain, and Calculated CMRR

V_1 (Sweep)	DC Gain [dB]	CM Gain [dB]	Calculated CMRR [dB]
600m	48.16	-4.61	52.77
700m	75.91	6.90	69.01
800m	78.16	-2.38	80.54
900m	78.52	-5.08	83.60
1.0	78.64	-6.47	85.11
1.1	78.63	-7.36	85.99
1.2	78.52	-7.95	86.47
1.3	78.28	-8.31	86.59
1.4	77.83	-8.47	86.30

5.3 Transient response before layout:

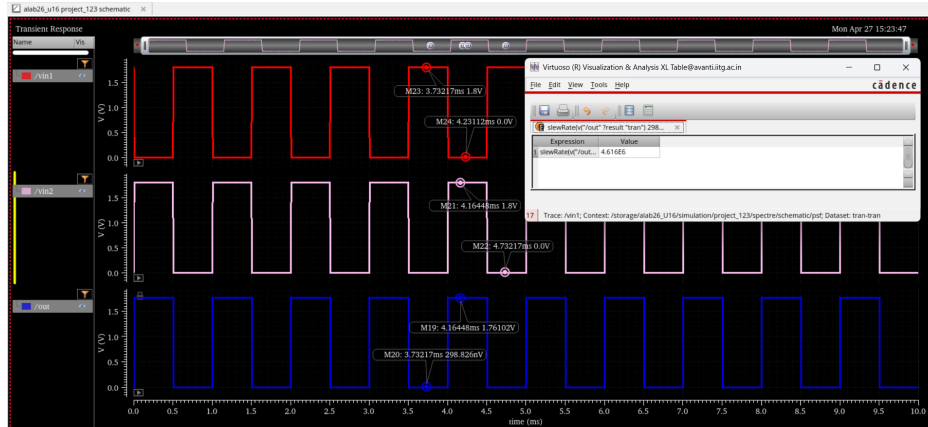


Figure 22: Slew rate Calculation

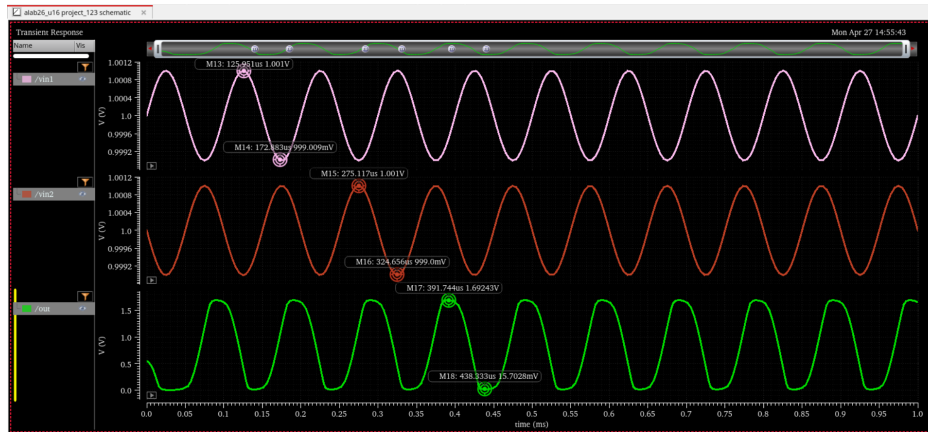


Figure 23: Output swing

*Key Design Insights

- Higher GBW $\rightarrow$ faster response
- Lower Power $\rightarrow$ better energy efficiency
- Smaller Area $\rightarrow$ lower cost and possibly higher yield in layout

5.4 Final Results before layout

Parameter	ICMR (V_1)	Theoretical	Practical
DC Gain (A_d)	0.7 V	75 dB	75.91 dB
	0.8 V	75 dB	78.16 dB
	0.9 V	75 dB	78.52 dB
	1.0 V	75 dB	78.64 dB
	1.1 V	75 dB	78.63 dB
	1.2 V	75 dB	78.52 dB
	1.3 V	75 dB	78.28 dB
	1.4 V	75 dB	77.83 dB
CM Gain (A_{cm})	0.7 V	-15 dB	6.90 dB
	0.8 V	-15 dB	-2.38 dB
	0.9 V	-15 dB	-5.08 dB
	1.0 V	-15 dB	-6.47 dB
	1.1 V	-15 dB	-7.36 dB
	1.2 V	-15 dB	-7.95 dB
	1.3 V	-15 dB	-8.31 dB
	1.4 V	-15 dB	-8.47 dB
CMRR	0.7 V	90 dB	69.01 dB
	0.8 V	90 dB	80.54 dB
	0.9 V	90 dB	83.60 dB
	1.0 V	90 dB	85.11 dB
	1.1 V	90 dB	85.99 dB
	1.2 V	90 dB	86.47 dB
	1.3 V	90 dB	86.59 dB
	1.4 V	90 dB	86.30 dB
Slew Rate	-	88 V/ μ s	4.616 V/ μ s
Output Swing	-	0.12 V to 1.62 V	0.016 V to 1.69 V
Power	-	0.7 mW	0.763 mW

6 Layout and post layout simulations

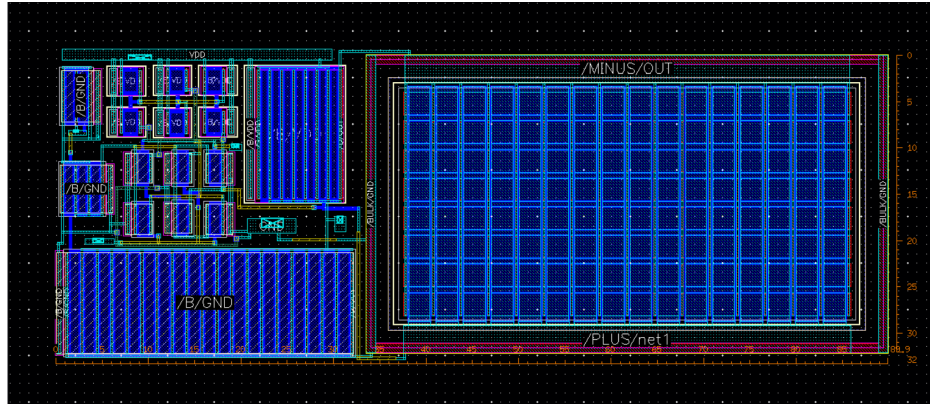


Figure 24: Layout design of two stage opamp

For the layout we have employed the common centroid method for the differential pair mosfets and the current mirror mosfets. We have used pattern of

A B A

B A B

for our design considering 3 multiplier for these each mosfets

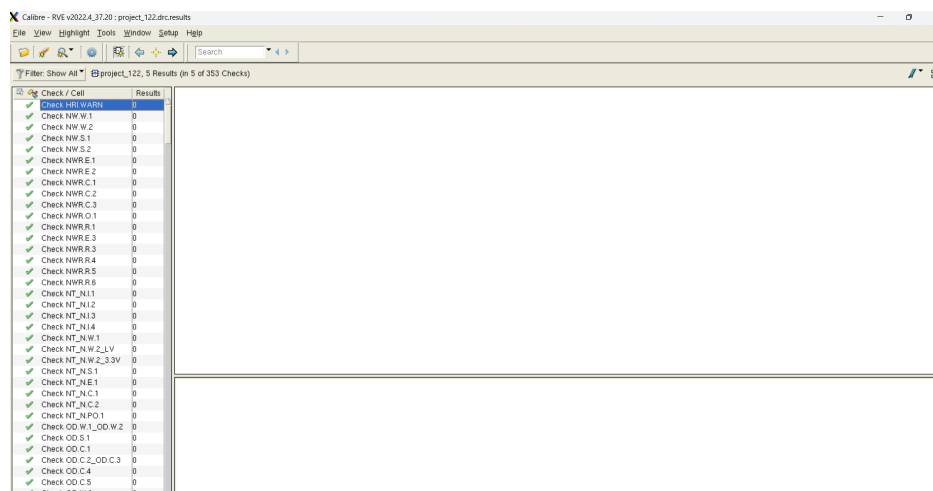


Figure 25: DRC report for layout

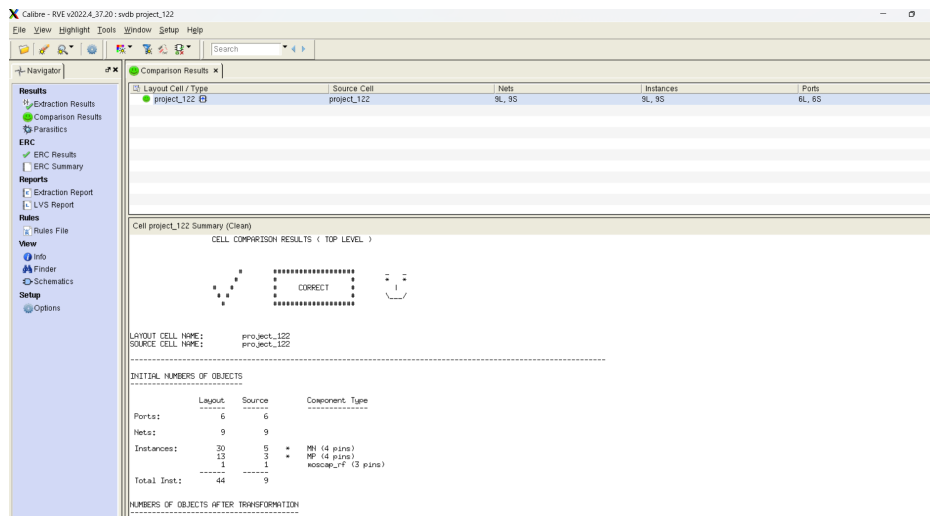


Figure 26: LVS report for layout

6.1 AC analysis after Parasitic extraction after layout:

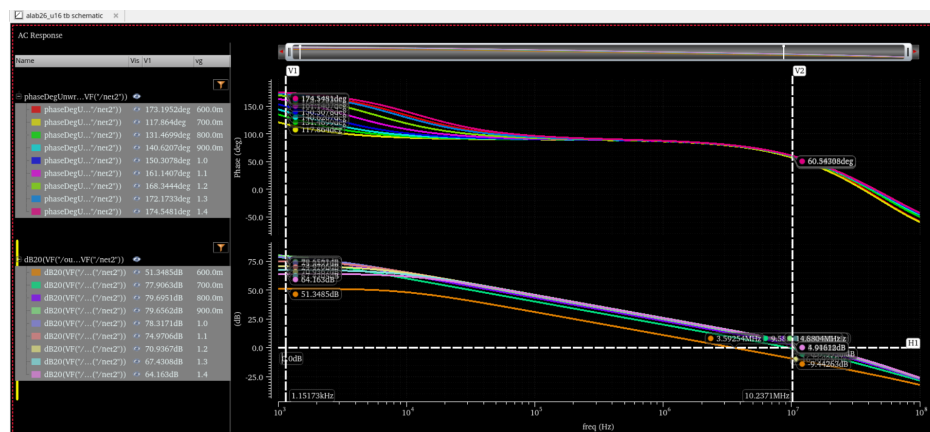


Figure 27: AC response of the amplifier showing gain and phase characteristics. For input voltage range from 0.6V to 1.4V

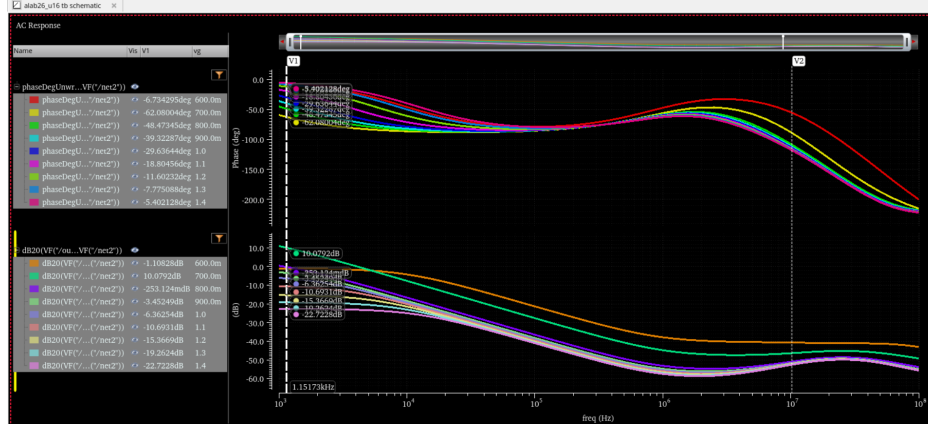


Figure 28: AC response of the amplifier showing common mode gain and phase characteristics. For input voltage range from 0.6V to 1.4V

Table 3: Differential Gain, Common-Mode Gain, and Calculated CMRR with parasitics after layout

V_1 (Sweep)	DC Gain [dB]	Common-Mode Gain [dB]	Calculated CMRR [dB]
600m	51.35	-1.11	52.46
700m	77.91	10.08	67.83
800m	79.70	-0.25	79.95
900m	79.66	-3.45	83.11
1.0	78.32	-6.36	84.68
1.1	74.97	-10.69	85.66
1.2	70.94	-15.37	86.31
1.3	67.43	-19.26	86.69
1.4	64.16	-22.72	86.88

6.2 Transient with parasitics after layout:

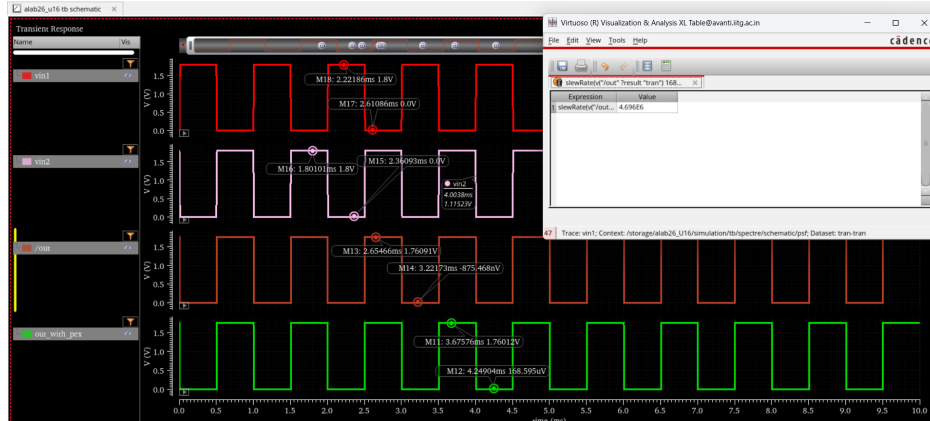


Figure 29: Slew rate Calculation

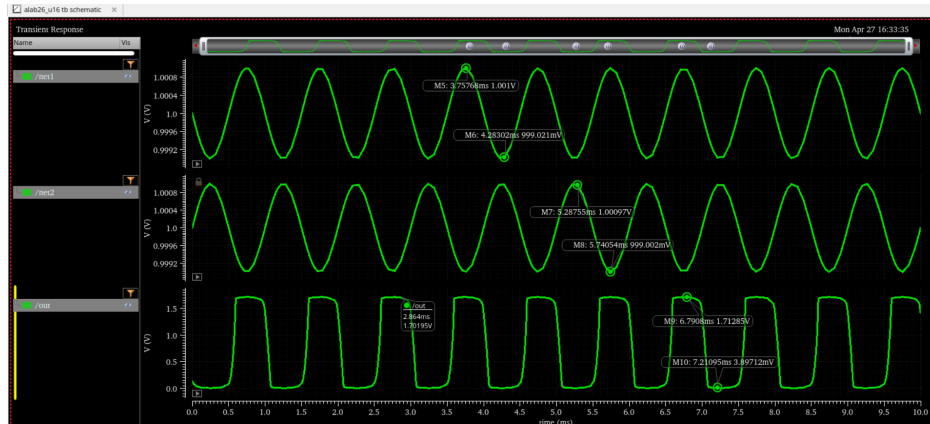


Figure 30: Output swing

6.3 Final Results before layout

Parameter	ICMR (V_1)	Theoretical	Practical
DC Gain (A_d)	0.7 V	75 dB	77.91 dB
	0.8 V	75 dB	79.70 dB
	0.9 V	75 dB	79.66 dB
	1.0 V	75 dB	78.32 dB
	1.1 V	75 dB	74.97 dB
	1.2 V	75 dB	70.94 dB
	1.3 V	75 dB	67.43 dB
	1.4 V	75 dB	64.16 dB
CM Gain (A_{cm})	0.7 V	-15 dB	10.08 dB
	0.8 V	-15 dB	-0.25 dB
	0.9 V	-15 dB	-3.45 dB
	1.0 V	-15 dB	-6.36 dB
	1.1 V	-15 dB	-10.69 dB
	1.2 V	-15 dB	-15.37 dB
	1.3 V	-15 dB	-19.26 dB
	1.4 V	-15 dB	-22.72 dB
CMRR	0.7 V	90 dB	67.83 dB
	0.8 V	90 dB	79.95 dB
	0.9 V	90 dB	83.11 dB
	1.0 V	90 dB	84.68 dB
	1.1 V	90 dB	85.66 dB
	1.2 V	90 dB	86.31 dB
	1.3 V	90 dB	86.69 dB
	1.4 V	90 dB	86.88 dB
Slew Rate	-	88 V/ μ s	4.696 V/ μ s
Output Swing	-	0.12 V to 1.62 V	0.0039 V to 1.71 V
Power	-	0.7 mW	0.78 mW

7 Conclusion

The design and physical implementation of the two-stage operational amplifier have been successfully completed. Based on the analysis of both pre-layout and post-layout results, the following conclusions can be drawn:

- **Gain and Stages:** To achieve the required gain, a two-stage op-amp structure was used. The first stage provides high gain with optimized transconductance (g_m), while the second stage ensures sufficient output swing and additional gain.
- **Bandwidth and Phase Margin:** Miller compensation using a capacitor (C_C) was successfully implemented to ensure a stable phase margin of 60° . The use of a nulling resistor allowed for better control over the Right-Half Plane (RHP) zero, enhancing stability.
- **Design Methodologies:** The g_m/I_D methodology was utilized to account for second-order effects, ensuring that the transistor sizing closely aligned with the required performance specifications in the simulator.
- **Layout and Parasitic Extraction:** A physical layout of the op-amp was developed, followed by **Parasitic Extraction (PEX)**. Post-layout simulations revealed the following:
 - **Gain Reduction:** A marginal decrease in DC gain was observed compared to the schematic simulation. This is attributed to parasitic resistances and increased output conductance (g_{ds}) caused by layout-dependent effects.
 - **Power Consumption:** A small increase in total power dissipation was noted, primarily due to the addition of parasitic capacitances and interconnect resistances that slightly shift the bias points.
 - **Performance Robustness:** Despite these variations, the post-layout results remained within the acceptable design margin, confirming the robustness of the initial schematic design.
- **Trade-offs:** The inclusion of an integrated resistor offered better power efficiency and smaller transistor sizes at the cost of increased layout

complexity and area. The choice between resistor-based and resistor-less designs depends on the specific trade-offs between stability, power, and IC fabrication constraints.

In conclusion, the design successfully meets the specified requirements. The transition from schematic to layout verified that the amplifier maintains its performance under realistic physical conditions, effectively preparing the design for potential hardware implementation.